WORKED SOLUTION

STEP III 2006, Question 7

Pure • First-Order Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q7

The key idea

Both differential equations are quadratics in $\mathrm{d}y/\mathrm{d}x$. Rather than attacking them as differential equations straight away, we solve each quadratic for the derivative, which splits the equation into two ordinary first-order equations. Hyperbolic identities make the roots collapse to something simple, and the conditions at $x = 0$ tell us which root to follow.

Solution

Part (i)

By the quadratic formula,

$$u = -\sinh x \pm \sqrt{\sinh^2 x + 1} = -\sinh x \pm \cosh x,$$

since $\sinh^2 x + 1 = \cosh^2 x$ and $\cosh x > 0$. Using $\cosh x + \sinh x = \mathrm{e}^x$ and $\cosh x - \sinh x = \mathrm{e}^{-x}$, the two roots are

$$u = \mathrm{e}^{-x} \qquad \text{or} \qquad u = -\mathrm{e}^x.$$

The differential equation is this quadratic with $u = \mathrm{d}y/\mathrm{d}x$, so at every x we must have

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{-x} \qquad \text{or} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\mathrm{e}^x.$$

The first root is always positive and the second always negative, so they never coincide. A solution with a continuous derivative therefore cannot jump from one root to the other, and the condition $\mathrm{d}y/\mathrm{d}x > 0$ at $x = 0$ picks out the first one throughout. Integrating, $y = -\mathrm{e}^{-x} + c$, and $y = 0$ at $x = 0$ gives $c = 1$:

$$y = 1 - \mathrm{e}^{-x}.$$

Part (ii)

Write $p = \mathrm{d}y/\mathrm{d}x$. If $\sinh y = 0$, that is $y = 0$, then the equation reduces to $2p = 0$. This is how $y \equiv 0$ comes to be a solution, and it is the one we are told to set aside. Where $y \neq 0$, the quadratic formula gives

$$p = \frac{-1 \pm \sqrt{1 + \sinh^2 y}}{\sinh y} = \frac{-1 \pm \cosh y}{\sinh y}.$$

With the half-angle forms $\cosh y - 1 = 2\sinh^2 \frac{y}{2}$, $\cosh y + 1 = 2\cosh^2 \frac{y}{2}$ and $\sinh y = 2\sinh \frac{y}{2} \cosh \frac{y}{2}$, the two roots are

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tanh \frac{y}{2} \qquad \text{or} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\coth \frac{y}{2}.$$

These never coincide, because their product is -1, whereas equal roots would have product $\tanh^2 \frac{y}{2} \geq 0$.

Ask yourself Which of the two equations can give a solution through the origin that is not identically zero?

The first equation, $y' = \tanh \frac{y}{2}$, has the solution $y \equiv 0$. Its right-hand side is a smooth function of y, so through $(0, 0)$ that is its only solution: this is the trivial solution again. The second equation is singular at $y = 0$, where $|\mathrm{d}y/\mathrm{d}x| \to \infty$, so its solution curve can meet $y = 0$ with a vertical tangent. That is the one we want. Separating the variables,

$$\int \tanh \frac{y}{2} \, \mathrm{d}y = -\int \mathrm{d}x \quad \Longrightarrow \quad 2 \ln \cosh \frac{y}{2} = -x + C.$$

Putting $y = 0$ at $x = 0$ gives $C = 0$, so $\cosh^2 \frac{y}{2} = \mathrm{e}^{-x}$. Since $\cosh y = 2\cosh^2 \frac{y}{2} - 1$,

$$\cosh y = 2\mathrm{e}^{-x} - 1.$$

We can check this directly. Differentiating gives $\sinh y \, y' = -2\mathrm{e}^{-x} = -(\cosh y + 1)$, which is exactly $y' = -\coth \frac{y}{2}$. This works for both signs of y, so the whole curve, upper and lower halves, is a solution. As $\cosh y \geq 1$, the curve exists only for $x \leq 0$. It is symmetric in the x-axis. Near the origin $1 + \frac{1}{2}y^2 \approx 1 - 2x$, so the curve looks like the parabola $y^2 = -4x$ there.

Asymptotes. As $x \to -\infty$ we have $\cosh y \to \infty$, so $|y| \to \infty$. Write $w = 2\mathrm{e}^{-x} - 1$. Then

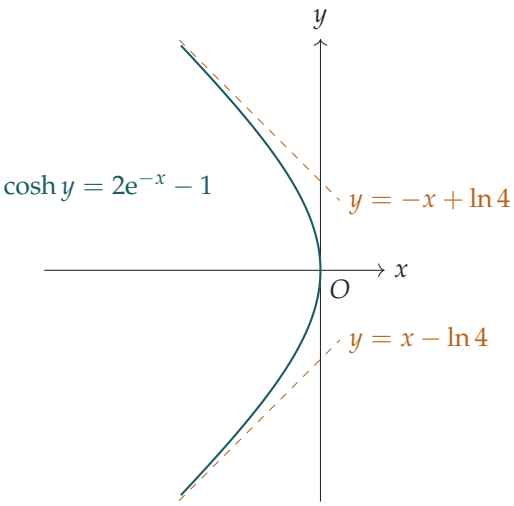
$$|y| = \operatorname{arcosh} w = \ln\bigl(w + \sqrt{w^2 - 1}\bigr),$$

so

$$|y| - (-x + \ln 4) = \ln \frac{w + \sqrt{w^2 - 1}}{4\mathrm{e}^{-x}} = \ln\Bigl(\frac{w}{4\mathrm{e}^{-x}}\Bigl(1 + \sqrt{1 - w^{-2}}\Bigr)\Bigr).$$

As $x \to -\infty$, $w/(4\mathrm{e}^{-x}) = \frac{1}{2} - \frac{1}{4}\mathrm{e}^{x} \to \frac{1}{2}$ and $1 + \sqrt{1 - w^{-2}} \to 2$, so the argument of the logarithm tends to 1 and the difference tends to 0. Hence the upper half of

the curve approaches $y = -x + \ln 4$ and, by symmetry, the lower half approaches $y = -(-x + \ln 4)$. The asymptotes are $y = \pm(-x + \ln 4)$, as required. (For $x \to 0^-$ the curve stays bounded, so these are the only asymptotes.)



Checks

- In (i), $y = 1 - \mathrm{e}^{-x}$ has $y' = \mathrm{e}^{-x}$, and $\mathrm{e}^{-2x} + 2\mathrm{e}^{-x}\sinh x - 1 = \mathrm{e}^{-2x} + (1 - \mathrm{e}^{-2x}) - 1 = 0$.
- In (ii), stepping $y' = -\coth\frac{y}{2}$ numerically from near the origin out to $x = -6$ gives $y \approx 7.38505$, matching $\operatorname{arcosh}(2\mathrm{e}^6 - 1)$. At $x = -10$ the gap between the curve and $-x + \ln 4$ is about 2×10^{-5}.

Common slips

- Taking the root $-\sinh x - \cosh x = -\mathrm{e}^x$ in (i): its derivative is negative at $x = 0$.
- Picking $y' = \tanh\frac{y}{2}$ in (ii) because it "behaves well" at $y = 0$. Through the origin it gives only $y \equiv 0$.
- Forgetting the lower half of the curve, which has $y < 0$ and is needed for the second asymptote.

Beyond STEP

An equation that is not linear in y', like these, is a *first-order implicit* equation. Solving for y' splits it into several branches. Part (ii) also shows that uniqueness of solutions can fail: at $y = 0$ the right-hand side $-\coth\frac{y}{2}$ is not Lipschitz, which is why a non-trivial solution can start from the same point as $y \equiv 0$. Boyce and DiPrima, *Elementary*

Differential Equations, covers existence and uniqueness theorems and when they apply.

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